

## P8109 Spring 2026 Homework #5

Due on April 30 at 11:59pm

1. From a lot with an unknown fraction of defective items  $p$ , we select a simple random sample of  $n$  elements. Suppose that  $p$  is a random variable with density

$$f(p) = \begin{cases} 1, & 0 \leq p \leq 1, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the posterior probability  $P(p \leq 0.1)$  given that 2 out of 30 items are defective. You may have to use the beta distribution

$$f(p \mid \alpha, \beta) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} p^{\alpha-1} (1-p)^{\beta-1}, \quad 0 < p < 1,$$

and a software package to calculate the posterior probability.

2. Suppose the result  $X$  of an IQ test is normally distributed with unknown mean  $\theta$  and variance 100. If the prior distribution of  $\theta$  is  $N(100, 225)$  and the observed score is  $x = 115$ , show that:
  - (a) the posterior distribution of  $\theta$  is  $N(110.38, 69.23)$ ;
  - (b) the Bayes factor for testing

$$H_0 : \theta \leq 100 \quad \text{vs.} \quad H_1 : \theta > 100$$

is 0.12.

3. An experiment started with  $n = 20$  rabbits. Let  $X$  be the number that survived and let  $p$  be the true probability of survival for each rabbit. Then  $X \sim \text{Binomial}(20, p)$ . Suppose that 15 rabbits survived and that  $p$  has a beta prior with parameters  $(\alpha, \beta) = (3, 5)$ , i.e.,

$$f(p \mid 3, 5) = \frac{\Gamma(8)}{\Gamma(3)\Gamma(5)} p^2 (1-p)^4, \quad 0 < p < 1.$$

Give your answers to two decimal places.

- (a) Compute the posterior Bayes estimator of  $p$ .
- (b) Obtain the maximum likelihood estimator of  $p$ .
- (c) How would your answer in (a) change if the prior of  $p$  were uniform on  $(0, 1)$ ?

- (d) Suppose now that 150 rabbits survived out of 200. Compare the MLE of  $p$ , the posterior Bayes estimator under a Beta(3, 5) prior, and the posterior Bayes estimator under a uniform prior.
4. Let the conditional density of  $X$  given  $\theta$  be gamma:

$$f(x | \theta) = \frac{\theta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\theta x}, \quad x > 0, \alpha > 0, \theta > 0,$$

and let the prior density of  $\theta$  be

$$\pi(\theta) = \frac{\gamma^\beta}{\Gamma(\beta)} \theta^{\beta-1} e^{-\gamma\theta}, \quad \theta > 0, \beta > 0, \gamma > 0.$$

That is,

$$X | \theta \sim \text{Gamma}(\alpha, 1/\theta), \quad \theta \sim \text{Gamma}(\beta, 1/\gamma).$$

Obtain the unconditional density of  $X$ .

5. Consider a parameter  $\theta$  with prior density

$$\pi(\theta) = \gamma\beta^\gamma \theta^{-(\gamma+1)}, \quad \theta \geq \beta,$$

where  $\beta > 0$  and  $\gamma > 2$ .

- Find the prior mean of  $\theta$ .
- Given observations  $Y_1 = y_1, \dots, Y_n = y_n$  from a Uniform(0,  $\theta$ ) distribution, find the posterior mean of  $\theta$ .
- Show that the estimator

$$\hat{\theta} = \frac{\gamma + n}{\gamma + n - 1} \max\{z, \beta\}, \quad z = \max(y_1, \dots, y_n),$$

is admissible under squared error loss.

6. It is often claimed that traveling by plane is safer than traveling by car since the respective probabilities of a crash are approximately  $5 \times 10^{-6}$  and  $10^{-2}$ . Even if these probabilities are accurate, is an airplane really much safer than a car?

|          | Travel by plane | Travel by car |
|----------|-----------------|---------------|
| Crash    | $10^6$          | 100           |
| No crash | 5               | 0             |

Above is the loss values associated with each decision (i.e., to travel in a plane vs. a car). Which decision is associated with a smaller risk?

7. Consider the following table of risks. The parameter space is  $\Theta = \{\theta_1, \theta_2, \theta_3, \theta_4\}$  with prior probabilities  $\{1/8, 3/8, 1/4, 1/4\}$ . The decision space is  $\{d_1, d_2, d_3\}$ .

|            | $d_1$ | $d_2$ | $d_3$ |
|------------|-------|-------|-------|
| $\theta_1$ | 0     | 2     | 3     |
| $\theta_2$ | 1     | 0     | 2     |
| $\theta_3$ | 3     | 4     | 0     |
| $\theta_4$ | 1     | 2     | 0     |

- (a) Find the Bayes risk of each decision.
  - (b) Which decision is Bayes?
  - (c) Obtain the minimax decisions.
8. Consider a single observation  $X \sim \text{Bernoulli}(\theta)$ . Suppose we estimate  $\theta$  using a Bayesian estimator  $d(X)$  such that

$$d(1) = d_1, \quad d(0) = d_2,$$

with loss function

$$L(d, \theta) = (d - \theta)^2.$$

- (a) Write down the risk function  $R_d(\theta)$ .
- (b) Show that the minimax decision satisfies

$$d_1 = \frac{3}{4}, \quad d_2 = \frac{1}{4}.$$